

Chapter 8: Direct Products

⊕ Gallian's notation

Normal notation: $\times$.

G_1 and G_2 groups.

$G_1 \times G_2$ is the Cartesian product, i.e.,

$$\{(g_1, g_2) : g_1 \in G_1, g_2 \in G_2\}.$$

This has the group structure defined by

$$\begin{aligned} (g_1, g_2) \cdot (g'_1, g'_2) \\ = (g_1 g'_1, g_2 g'_2). \end{aligned}$$

$|G_1 \times G_2| = |G_1| |G_2|$. This is called the **direct product** of groups.

If G_1 and G_2 are abelian,
 $G_1 \times G_2$ abelian and we
also use the notation
 $G_1 \oplus G_2$ (direct sum),
especially when group
operation written as
addition. The unit
element is $(0, 0)$ when
we write group operation
as addition, but the first
 0 is in G_1 and the second
is in G_2 . We write $(1, 1)$
for the unit element if
group is multiplication.

Examples

1. $G_1 = G_2 =$ cyclic group $\mathbb{Z}/2$ (with addition as group operation). Then $G_1 \oplus G_2$ consists of $(0, 0), (0, 1), (1, 0), (1, 1)$ and is the **Klein 4-group**. This is the unique noncyclic group of order 4 (up to isomorphism).

In any direct product $G_1 \times G_2$, G_1 is embedded

as $G_1 \times \{e_{G_2}\}$ and G_2 is embedded as $\{e_{G_1}\} \times G_2$, and these two subgroups commute with each other.

Recognition principle

A group G is isomorphic to $H \times K \iff G$ contains subgroups isomorphic to H and K , respectively, which commute with each other and generate all of G .

Example 2

$G = (\mathbb{Z}/n, +)$ where

$n = mk$ and $\gcd(m, k) = 1$.
Then $G \cong \mathbb{Z}/m \times \mathbb{Z}/k$.

G is cyclic and consists
of $\bar{0}, \bar{1}, \dots, \overline{n-1}$, where
— denotes equivalence
class mod n . Define

$$\varphi: G \rightarrow \mathbb{Z}/m \times \mathbb{Z}/k$$

by $\varphi(\bar{x}) = (x \bmod m, x \bmod k)$.

This is a group homomorphism
since it respects addition.

φ is 1-to-1 since if

$$\begin{aligned} x &\equiv y \pmod{m} \text{ and} \\ x &\equiv y \pmod{k} \end{aligned}$$

then. since $\gcd(m, k) = 1$,
we have u and v
with $um + vk = 1$,

$$\text{and so } \begin{cases} x = y + mw \\ x = y + kz \end{cases}$$

So $x - y$ is divisible by m
and divisible by k , so
it's divisible by both.

But $m \cdot k = n$ and m and
 k are relatively prime, so
divisible by n . So we
have an isomorphism.

Similar thing happens

with multiplicative groups. Again assume $n = mk$ and $\gcd(m, k) = 1$. We get a map

$$(\mathbb{Z}/n^{\times}, \times) \rightarrow (\mathbb{Z}/m^{\times}, \times) \times (\mathbb{Z}/k^{\times}, \times)$$

by same formula

$$\bar{x} \mapsto (x \bmod m, x \bmod k).$$

This respects multiplication, and it's still one-to-one, so we get an isomorphism

$$(\mathbb{Z}/n)^{\times} \longrightarrow (\mathbb{Z}/m)^{\times} \times (\mathbb{Z}/k)^{\times}$$

(Theorem 8.3 in Gallian.)

Example

$$(\mathbb{Z}/12)^{\times} = \{1, 5, 7, 11\}$$

$$12 = 3 \cdot 4$$

$$(\mathbb{Z}/4)^{\times} = \{1, 3\}$$

$$(\mathbb{Z}/3)^{\times} = \{1, 2\}$$

$$5^2 = 25 = 1 \pmod{12}$$

$$7^2 = 49 = 1 \pmod{12}$$

$$11^2 = 121 = 1 \pmod{12}$$

So $(\mathbb{Z}/12)^{\times}$ is an abelian group of 4 elements,

3 of which have order 2.

The map $x \mapsto (x \bmod 4, x \bmod 3)$

sends $5 \mapsto (1, 2)$

$7 \mapsto (3, 1)$

$11 \mapsto (3, 2)$

get an isomorphism

$$(\mathbb{Z}/12)^\times \cong (\mathbb{Z}/4)^\times \times (\mathbb{Z}/3)^\times.$$

What's the point?

Idea is that direct products give a way to split groups into smaller pieces (that actually commute with each other). If the smaller pieces don't commute

with each other, you don't get a direct product, but sometimes you get what's called a **semidirect** product.

This is a case where $G = H \cdot K$ with H and K subgroups and $H \cap K = \{e\}$.

Example $S_3 = H \cdot K$,
where $H = \{1, (123), (132)\}$
 $K = \{1, (12)\}$. H and K

do not commute with each other.

Theorem 8.2 Let

G and H be finite cyclic groups. Then $G \times H$

is cyclic $\iff |G|$ and $|H|$ are relatively prime.

Proof $\Leftarrow$ If G has

order m , H has order k , and $\gcd(m, k) = 1$, we saw that $\mathbb{Z}/mk \cong \mathbb{Z}/m \times \mathbb{Z}/k$.

$\Rightarrow$ If $G \times H$ is cyclic,

Let (a, b) be a cyclic generator. Then $\langle a \rangle = G$ and $\langle b \rangle = H$, so G and H are cyclic.